

Response to the Meta-Reviewer

We sincerely thank the Meta-Reviewer for carefully considering our manuscript and for providing valuable guidance through the review process. We also appreciate the constructive feedback provided by the reviewers.

In this document, we provide clarifications and responses to the concerns raised during the review process. We have carefully considered the feedback and have made every effort to address the reviewers' concerns with additional explanations, experimental details, and results where appropriate.

Major points by Reviewer rBVo

1. No expert ground truth; the model learns the pipeline's own labels.

Agreed. The proposed work agrees with the rule-based logic, not correctness of the notation. We don't claim of expert-verified accuracy. The paper claims the output only as a first draft for expert review and Section 6 shows this limitation.

Validation by Labanotation experts and Karagattam gurus is the necessary next step and is not claimed here. During the development of the work, we also discussed the approach and generated representations with dance gurus associated with Kalai Kaviri, which provided domain-oriented feedback; however, we acknowledge that these discussions do not constitute an independent expert annotation or formal validation study.

2. Phrase segmentation is evaluated on a single case study, with no expert boundaries or metrics.

Agreed. We do not report a quantitative segmentation metric or claim quantified segmentation accuracy. The purpose of this analysis was to demonstrate the possibility of identifying non-uniform movement phrases and their temporal structure using the proposed self-similarity-based approach. We acknowledge that systematic evaluation against expert-annotated temporal boundaries across the dataset is an important direction for future work.

3. The dataset is very small: 6 dancers, 96 studio videos.

Agreed. The 96 videos ($16 \text{ adavus} \times 6 \text{ dancers} \times 3 \text{ speeds}$) are a small set. Dancer-disjoint 3-fold CV ensures that none of the dancer appeared in both training and testing (Sec. 3, 4.8), but this does not remove the limit on diversity. Generalisation beyond this setting is not shown in the paper. Such broader validation is an important direction for future work.

4. Depth direction, support, foot contact, effort and facing cannot be reliably inferred from 2D.

Agreed. We accept that several notation attributes are difficult to infer reliably from monocular 2D pose. In our results, arm direction (LArmDir: 53.4%, RArmDir: 53.3%) and foot-contact attributes (47.1–53.8%) shows comparatively lower performance than torso-level and torso-direction attributes. We therefore do not interpret these lower-performing columns as reliable automatic notation.

The reported results are intended to say these limitations rather than imply that all notation attributes can be reliably recovered from a single camera. We identify multi-view/depth-aware sensing as an important direction for improving these attributes.

5. Shoulder-based rotation normalisation may remove orientation needed for direction.

The shoulder-based rotation normalization in our pipeline was intended primarily to reduce minor in-plane alignment variations between the performer and camera, rather than to model large body. rotations.

We clarify that the controlled front-facing recording setup does not imply that Karagattam performers do not turn during performance. Karagattam may involve turns to different spatial directions; however, for this initial monocular dataset, all adavus were recorded with the performer facing the camera to maintain a consistent viewpoint for pose extraction and movement analysis. Based on discussions with a Karagattam dance guru associated with Kalai Kaviri, such turns primarily change the performer's spatial orientation/direction, while the underlying adavu movement remains the same.

6. Composite selection is worse than Uniform on silhouette, coverage, PCK and velocity; only DB improves.

Agreed that the advantage is weak. In Table 3 Composite wins only on DB (1.506 vs 1.636 for Uniform). Uniform is better on silhouette, coverage and PCK@0.20. Composite targets stable, readable postures rather than clustering metrics, but our results do not show an overall advantage, and we do not claim one.

The composite score was designed to prioritize stable and readable postures by jointly considering movement-related criteria.

7. K is chosen by maximising silhouette, then silhouette is reported as evidence (selection bias).

Agreed that silhouette is not independent evidence. The remaining indicators, DB (0.800), intra/inter ratio (0.205) and LOO-ARI (0.734), were not used to select K, though they are computed on the same clustering.

8. Conv1D-BiLSTM gets 30.2% adavu accuracy vs 89.6% for Random Forest.

Agreed. The adavu classification head is an auxiliary output, and its performance (30.2%) is substantially lower than the Random Forest baseline (89.6%). We believe small data relative to model capacity the reason, but we have not tested that.

We do not claim that the proposed deep model is effective for adavu classification. The primary objective of the model is per-frame Laban-inspired notation prediction, while adavu recognition was included as an auxiliary output rather than as a main contribution.

9. Per-column accuracy is inadequate under class imbalance; need macro-F1, per-class recall, confusion matrices and majority baselines.

Column	Macro-F1	Majority Baseline	AMacro-F1
Support	0.7270	0.1823	+0.545
GestureSide	1.0000	0.3708	+0.629
GestureDir	1.0000	0.4988	+0.501
LArmDir	0.5782	0.2231	+0.355
RArmDir	0.5484	0.1977	+0.351
FootContactL/R	~0.53	~0.25	+0.25-0.30
HeadDir	0.2628	0.2628	0.000
SupportL Level	0.3862	0.3878	- 0.002
RArmLevel	0.4924	0.4924	0.000

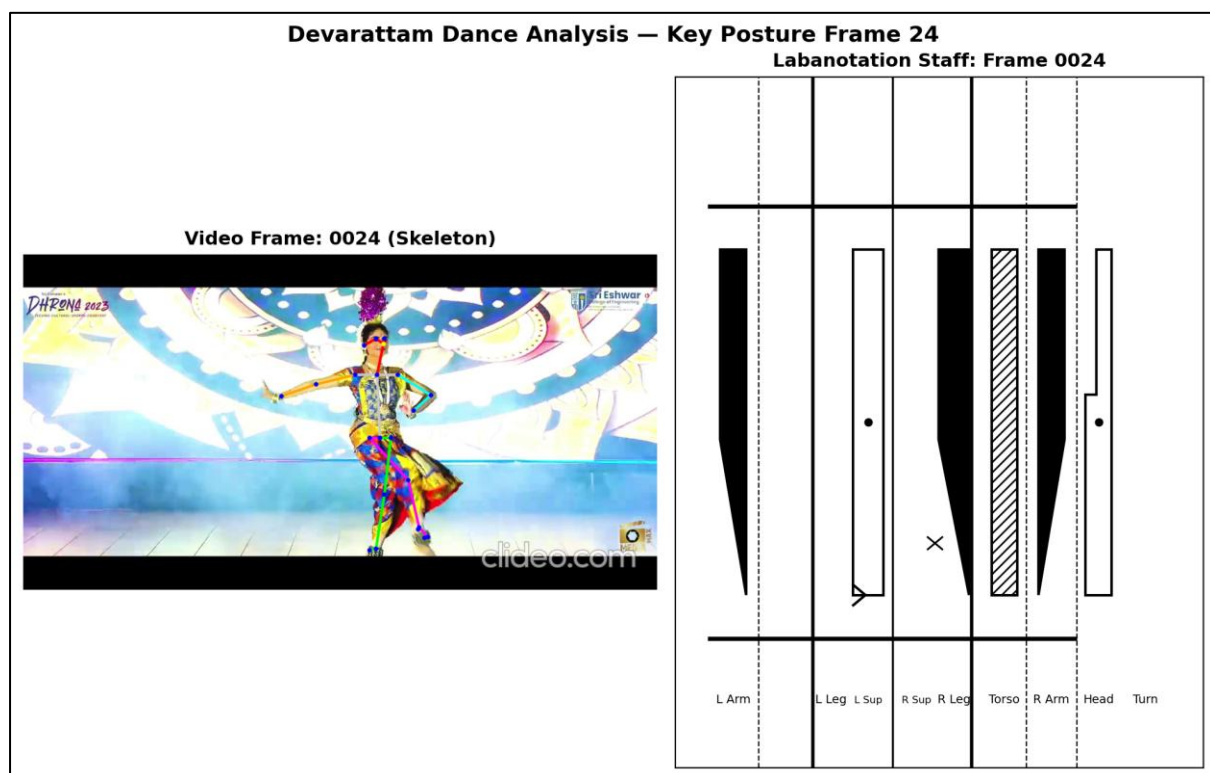
We agree that accuracy alone can be misleading under class imbalance. We therefore additionally evaluated macro-F1 against a majority-class baseline. Support achieves **0.727 vs. 0.182**, and LArmDir **0.578 vs. 0.223**, demonstrating gains beyond the majority baseline for these columns. However, some columns tend to collapse to the majority class, reflecting insufficiently balanced samples. Thus, we agree that individual 100% accuracy values should not be interpreted as expert-validated Labanotation accuracy. We will clarify this limitation and include the additional macro-F1/baseline analysis in the revised manuscript.

11. Robustness uses only 8–11 postures and no repeated trials.

Agreed on both points. Table 4 reports a single run (fixed seed 42) on one representative video with 8-11 postures, so seed variance is not quantified. The non-monotonic F1 (0.667 at 2 px, 0.778 at 5 px, 0.600 at 10 px) is consistent with small-sample variation, since one missed posture shifts F1 substantially at $n \approx 9$.

12. Only a static camera, clean background, single dancer and controlled lighting.

We agree that the reported dataset was collected under controlled conditions. However, the proposed pose-extraction stage uses OpenPose, which estimates human body keypoints based on detected human body appearance rather than relying on a fixed background or specific costume. To further examine its applicability beyond our controlled dataset, we additionally tested the pose-estimation pipeline on a publicly available Karagattam video from YouTube, containing a different background and performance setting. OpenPose successfully detected the required body keypoints, and the resulting pose-estimation output is provided as supplementary evidence. This demonstrates that the pose-extraction stage is not restricted to the studio background used for dataset collection. However, systematic evaluation under camera motion, severe occlusion, and multiple-dancer scenarios remains outside the scope of the present study.



13. No annotation agreement, dancer consent, data governance or community validation described.

We clarify that informed consent was obtained from all participating dancers for video recording and use of their performances for this research. We will include a sample consent form in the supplementary material to clarify the data-collection and research-use permissions. The videos were collected from a government aided dance institute, specifically for this study.

Regarding annotation agreement, our objective is to develop an automated first-pass Labanotation generation framework. The current labels are generated through the proposed rule-based mapping from extracted pose features; therefore, we do not claim that the reported classification accuracy represents agreement with independently expert-annotated Labanotation. Independent expert annotation and inter-annotator agreement would provide stronger validation and are planned as an important next stage of the work. We have clarified this distinction and the corresponding limitation in the revised manuscript.

VOLUNTEER CONSENT FORM
Dataset Collection – Karagattam

Labanotation generation
 Automated Labanotation generation
 Automated Labanotation generation

Volunteer Name: Mobile Number:

PURPOSE: Your video recording will be used as ground truth / reference data for the automated Labanotation generation. The data will not be used for any purpose outside this project.

I confirm the following:

- ☒ I am voluntarily participating in this video recording session.
- ☒ I understand the recording will be used only as data reference / ground truth data for the above research project.
- ☐ I have no objection to the research team using my movement recordings for the stated purpose.

 Signature of Volunteer

1/4/26
 Recording date

 Signature of Coordinator

 Name & Designation

This form will be retained by the research team for internal records.
 For queries contact (do)

Minor points (rBVo)

1. Eq. (2) is a distance but is called a similarity matrix.

$S_{ij} = \|f_i - f_j\|_2$ represents a pairwise distance matrix rather than a similarity matrix. We acknowledge that the terminology in the manuscript is imprecise. The same distance matrix is directly used by the checkerboard novelty computation for phrase-boundary detection, so this terminology issue does not affect the reported computation or results of the work.

2. The arctan2 description swaps horizontal and vertical.

Correct. The accurate text is: $\theta = \arctan2(\Delta y_{sh}, \Delta x_{sh})$, where Δx_{sh} and y_{sh} are the horizontal and vertical differences between the two shoulder joints. This is a wording error only. The computation is unchanged.

This is a typographical error in the description only. The implementation and all reported results are unaffected.

3. PSS thresholds and weights are manually set with no sensitivity analysis.

Agreed. The PSS weights (0.35, 0.25, 0.20, 0.20) are hand-set heuristics, chosen so that still, stable frames (velocity, acceleration) are favoured over reach and contraction. They were not optimised, and we claim no optimality. In the submitted paper only the minimum inter-posture distance was swept (Table 5: silhouette 0.268 at 10-12 frames, falling to 0.197 at 25 frames). The weights were not.

As additional evidence not in the paper, we ran a post-hoc sensitivity analysis on the Sabai Vanakkam video (baseline: 22 postures, silhouette 0.350). Each weight was perturbed by ± 0.05 and ± 0.10 and the vector renormalised (16 perturbations). Results:

<i>Perturbed weight</i>	<i>Δ</i>	<i>Postures</i>	<i>Silhouette</i>	<i>F1 vs baseline set</i>
<i>velocity</i>	<i>-0.10</i>	23	0.326	0.933
<i>velocity</i>	<i>+0.10</i>	23	0.349	0.978
<i>acceleration</i>	<i>-0.10</i>	22	0.337	1.000
<i>acceleration</i>	<i>+0.10</i>	22	0.348	0.955
<i>reach</i>	<i>-0.10</i>	25	0.347	0.936
<i>reach</i>	<i>+0.10</i>	22	0.343	0.955
<i>contraction</i>	<i>-0.10</i>	20	0.320	0.857
<i>contraction</i>	<i>+0.10</i>	23	0.296	0.889

Across all 16 perturbations the posture count ranged from 20 to 25, silhouette from 0.296 to 0.355, and F1 agreement with the baseline posture set from 0.857 to 1.000. Contraction was the most sensitive weight. The selected posture set is therefore reasonably stable to ± 0.10 changes, but this covers one video and one seed, and it measures the stability of the selected set, not its correctness. The representative-selection weights in Eq. (5) were not swept.

4. Rules and thresholds for support, foot contact, direction and effort are incomplete.

Agreed. [If you can, paste the actual thresholds here, e.g. the ankle-height margin for support, the direction bin angles and the effort cutoffs. Only include values from your code.]

Agreed. The submitted paper does not list the thresholds. All rules are deterministic geometric functions of the normalised 2D pose, with no learned parameters. The values from our implementation are below. Distances are relative to torso length.

Rule	Definition
Floor estimate	95th percentile of raw ankle-y over the video
Foot on ground	lowest of ankle/heel/toe within $0.22 \times$ torso length of the floor
Support (L/R/Both/None)	foot on ground per side; if both are down and $ l_drop - r_drop > 0.15$, support goes to the straighter leg
Foot contact	aerial: foot not on ground. whole: heel and toe both within 0.22. ball: toe within 0.22, heel above. heel: heel within 0.22, toe above
Support leg level	Low: knee $< 155^\circ$. High: knee $\geq 155^\circ$ and ball contact. Middle: knee $\geq 155^\circ$ and whole/heel contact
Knee flexion (0–4)	$\geq 168^\circ$, $158\text{--}168^\circ$, $148\text{--}158^\circ$, $138\text{--}148^\circ$, $< 138^\circ$
Gesture flexion (0–2)	$\geq 165^\circ$, $145\text{--}165^\circ$, $< 145^\circ$
Direction (8 bins)	"place" if vector magnitude < 0.12 ; otherwise 45° sectors, $\text{round}(\text{angle}/45) \bmod 8$
Level	High: $dy/\text{torso} > +0.35$. Middle: between. Low: ≤ -0.15
Torso tilt	place < 0.18 ; low > 0.45
Head tilt	place < 0.08 ; low > 0.22
Effort_Sudden / Effort_Free	continuous scores over a 5-frame window: smoothed acceleration magnitude, and $1/(1 + 10 \cdot \text{jerk})$, $\sigma = \text{window}/3$

Effort is used as a continuous input feature with no cutoff. These thresholds were set manually, in part following Kinetography Laban conventions [8], and were not tuned or validated against expert annotation or motion-capture reference. The deep model is trained on labels from these rules, so its accuracy depends on them.

5. "whatsapp_video_2.mp4" is not a suitable identifier.

Agreed. It is a representative Karagattam clip from the dataset; the filename is the internal identifier, and the paper should describe it as "the representative video".

6. Labels such as SuppL_Level and Gest._Flex need expanding.

Agreed. SuppL_Level = left-support level; SuppR_Flex = right-support knee flexion; Gest._Flex = gesture flexion ; GestureSide = side of the gesturing limb; LArmDir/RArmDir = left/right arm direction.

7. Figures 3 and 4 are too small to verify.

Agreed. Figures 3 and 4 are relatively small and may make some details difficult to inspect. We acknowledge that larger and clearer figures would improve readability and presentation of the pipeline and intermediate results.

8. Header "ICVGIP 2026, 2026, India" is incomplete

Agreed. The missing DOI/ISBN information is a bibliographic metadata issue and does not affect the technical content, methodology, or experimental results of the paper.

9.DOI/ISBN missing.

Agreed. The missing DOI/ISBN information is a bibliographic metadata issue and does not affect the technical content, methodology, or experimental results of the paper.

Reviewer hyVA

Q1. Video length, fps, BiLSTM sequence length, and the temporal window needed.

The videos were recorded at approximately 30 fps. As an example, the Sabai Vanakkam sequence contains 860 frames, corresponding to approximately 28.7 seconds. The videos have varying durations rather than a single fixed sequence length. The Conv1D-BiLSTM processes the frame-level feature sequences and provides temporal context in both forward and backward directions within the input sequence.

We did not perform a temporal-window ablation in the current study. Therefore, we cannot specify the minimum temporal window required for reliable Labanotation prediction. We acknowledge that systematically evaluating different temporal context lengths would be valuable for determining how much movement history is necessary for different notation attributes.

Q2. How were the 23 features determined? References? Meaningful from a dance perspective?

The 23 values are the per-frame input to the sequence model (Sec. 4.8): 8 joint angles (elbow, shoulder, hip, knee, left and right) and 15 movement descriptors. The effort, shape and space features follow Laban Movement Analysis concepts [10, 11]. The rest are standard kinematic and rhythm descriptors. The set was chosen by the authors to cover the notation columns being predicted. We ran no feature ablation and no expert review, so the features are motivated by LMA but their dance meaning has not been validated.

Q3. How is foreshortening handled without 3D?

It is not. Normalisation by torso length reduces scale and camera-distance effects but does not correct foreshortening. This is why arm direction and foot contact stay at about 53% and 47–54% (Table 7). A calibrated multi-view setup, as the reviewer suggests, would be the principled remedy.

Q4. Basis for the weights in Section 4.4.

We acknowledge that the weights used in Section 4.4 were hand-set and that no formal sensitivity analysis or optimization was performed. The four weights—velocity (0.35), acceleration (0.25), reach (0.20), and contraction (0.20)—were chosen heuristically to favour temporally stable and clearly held postures during keyframe selection. Accordingly, velocity and acceleration together receive a higher weight (0.60) than reach and contraction (0.40), based on the observation that clearer, more stable poses are preferable to frames captured during transitional or mid-swing movements.

These weights were informed by manual inspection of keyframe quality on trial videos rather than by a formal grid search or data-driven optimization. We agree that this design choice was not sufficiently explained in the manuscript. We also acknowledge that the selected weights are not necessarily optimal and may not generalize across different dance styles, performers, or tempos.

A systematic sensitivity analysis or data-driven weight optimization would provide stronger validation of the posture-selection strategy and is an important direction for future work.

Q5. Is 96 videos too few for a BiLSTM?

It is small, and the 30.2% adavu result is consistent with that, though we have not tested the cause. For per-frame notation the supervision is dense, and we used dropout, label smoothing, augmentation and dancer-disjoint CV. High scores on deterministic columns do not show generalisation beyond this dataset.

We sincerely appreciate the Meta-Reviewer's time, consideration, and constructive guidance. We hope that the clarifications and additional evidence provided in our responses adequately address the concerns raised during the review process.